

97/100 Arwah Faisal

homework 1

$$y' = f(x) g(y)$$

$$\frac{dy}{dx} = f(x) g(y)$$

$$\frac{dy}{g(y)} = \int f(x) dx$$

$$A = B + C$$

$$b. y' + P(x)y = Q(x) \times [e^{\int P(x) dx}]$$

$$y e^{\int P(x) dx} + P(x)y e^{\int P(x) dx} = Q(x) e^{\int P(x) dx}$$

$$\frac{d}{dx} [y e^{\int P(x) dx}] = Q(x) e^{\int P(x) dx}$$

$$y = \frac{\int Q(x) e^{\int P(x) dx}}{e^{\int P(x) dx}} + \frac{C}{e^{\int P(x) dx}}$$

$$c. y'' + b^2 y = 0 \quad * y = e^{rx}$$

$$(e^{rx})'' + b^2 e^{rx} = 0$$

$$e^{rx}(r^2 + b^2) = 0$$

$$r = \pm ib$$

$$y = C_1 \cos(bx) + C_2 \sin(bx)$$

$$d. y'' - b^2 y = 0 \quad * y = e^{rx}$$

$$(e^{rx})'' - b^2 e^{rx} = 0$$

$$e^{rx}(r^2 - b^2) = 0$$

$$r = \pm b$$

$$y = C_1 e^{bx} + C_2 e^{-bx}$$

$$e. y'' = b_0 + b_1 x + b_2 x^2$$

$$y' = b_0 x + \frac{b_1 x^2}{2} + \frac{b_2 x^3}{3} + C$$

$$y = \frac{b_0 x^2}{2} + \frac{b_1 x^3}{6} + \frac{b_2 x^4}{12} + C_1 x + C_2$$

$$f. \frac{1}{x} \frac{d}{dx} \left(x \frac{dy}{dx} \right) = b_0 + b_1 x + b_2 x^2$$

$$\frac{d}{dx} \left(x \frac{dy}{dx} \right) = \int (b_0 x + b_1 x^2 + b_2 x^3) dx$$

$$x \frac{dy}{dx} = \frac{b_0 x^2}{2} + \frac{b_1 x^3}{3} + \frac{b_2 x^4}{4} + C_1$$

$$\frac{dy}{dx} = \int \left(\frac{b_0 x}{2} + \frac{b_1 x^2}{3} + \frac{b_2 x^3}{4} + \frac{C_1}{x} \right) dx$$

$$y = \frac{b_0 x^2}{4} + \frac{b_1 x^3}{9} + \frac{b_2 x^4}{16} + C_1 \ln|x| + C_2$$

$$g. \frac{1}{x^2} \frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) + b^2 y = 0 \quad * y(x) = \frac{u(x)}{x} \rightarrow y' = (xu'(x) - u(x))/x^2$$

$$\frac{1}{x^2} \frac{d}{dx} [xu'(x) - u(x)] + \frac{b^2 u(x)}{x} = 0$$

$$\frac{1}{x^2} [u'(x) + xu''(x) - u'(x)] + \frac{b^2 u(x)}{x} = 0$$

$$u''(x) + b^2 u(x) = 0 \rightarrow u'(x) + b^2 = 0$$

$$u(x) = \sqrt{-b^2} = \pm bi$$

$$u(x) = C_1 \cos bx + C_2 \sin bx$$

$$y = \frac{C_1 \cos bx + C_2 \sin bx}{x}$$

$$h. \frac{1}{x^2} \frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) - b^2 y = 0 \quad * y(x) = \frac{u(x)}{x} \quad y' = (xu'(x) - u(x))/x^2$$

$$\frac{1}{x^2} \frac{d}{dx} [xu'(x) - u(x)] - \frac{b^2 u(x)}{x} = 0$$

$$\frac{1}{x^2} [u'(x) + xu''(x) - u'(x)] - \frac{b^2 u(x)}{x} = 0$$

$$u''(x) - b^2 u(x) = 0 \rightarrow u'(x) - b^2 = 0$$

$$u(x) = \pm b$$

$$u(x) = C_1 e^{bx} + C_2 e^{-bx} \rightarrow y = \frac{C_1 e^{bx} + C_2 e^{-bx}}{x}$$

$$i. y'' + b_1 y' + b_2 y = 0$$

$$(D^2 + b_1 D + b_2)y = 0$$

$$D = \frac{-b_1 \pm \sqrt{b_1^2 - 4b_2}}{2}$$

$$y = C_1 e^{\left(\frac{-b_1 + \sqrt{b_1^2 - 4b_2}}{2}\right)x} + C_2 e^{\left(\frac{-b_1 - \sqrt{b_1^2 - 4b_2}}{2}\right)x}$$

missing cases: $\Delta < 0$ & $\Delta = 0$
(-3)

$$j. y'' + 2xy' = 0 \quad * y' = u \quad u' = -2xu \rightarrow \frac{du}{u} = -2x dx \quad u = e^{-x^2 + C_1}$$

$$y'' = -2xy'$$

$$y' = e^{-x^2 + C_1} \rightarrow y = e^{C_1} \frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + C_2$$

$$y = C_1 \frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + C_2$$

$$k. \int \frac{d}{dx} \left[f(y) \frac{dy}{dx} \right] = 0$$

$$f(y) \frac{dy}{dx} = C_1$$

$$\int f(y) dy = \int C_1 dx$$

$$A = C_1 x + C_2$$

$$l. \frac{1}{x^2} \frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) = b_0 + b_1 x + b_2 x^2$$

$$\frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) = \int (b_0 x^2 + b_1 x^3 + b_2 x^4) dx$$

$$x^2 \frac{dy}{dx} = \frac{b_0 x^3}{3} + \frac{b_1 x^4}{4} + \frac{b_2 x^5}{5} + C_1$$

$$\frac{dy}{dx} = \int \left(\frac{b_0 x}{3} + \frac{b_1 x^2}{4} + \frac{b_2 x^3}{5} + \frac{C_1}{x^2} \right) dx$$

$$y = \frac{b_0 x^2}{6} + \frac{b_1 x^3}{12} + \frac{b_2 x^4}{20} - \frac{C_1}{x} + C_2$$